



Neuroeconomics Class Presentation Draft

Nature, 645, 991–1001

May 2026

- 1 Opening
- 2 Design
- 3 Methods
- 4 MouseResults
- 5 AIResults
- 6 Interpretation
- 7 Extension
- 8 Discussion

Opening

Why this paper matters for neuroeconomics

Core idea

Social decision-making may not live entirely inside one brain. During interaction, two agents may form a coupled neural system with measurable **shared neural dynamics**.

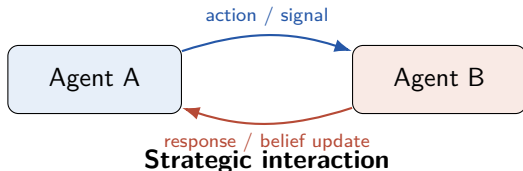
- The paper studies both **freely interacting mice** and **interacting AI agents**.
- The authors ask whether social interaction creates a real **shared neural structure**, not just matching sensory input or matching movement.
- That question is deeply neuroeconomic because many decisions are **strategic, reciprocal, and interdependent**.

Standard framing

- How does one brain process a social cue?
- How does one individual make a decision?
- The partner is often treated like the environment.

This paper's framing

- Two interacting agents form a feedback loop.
- My action changes your state, and your response changes mine.
- Neural activity may also become coupled across agents.

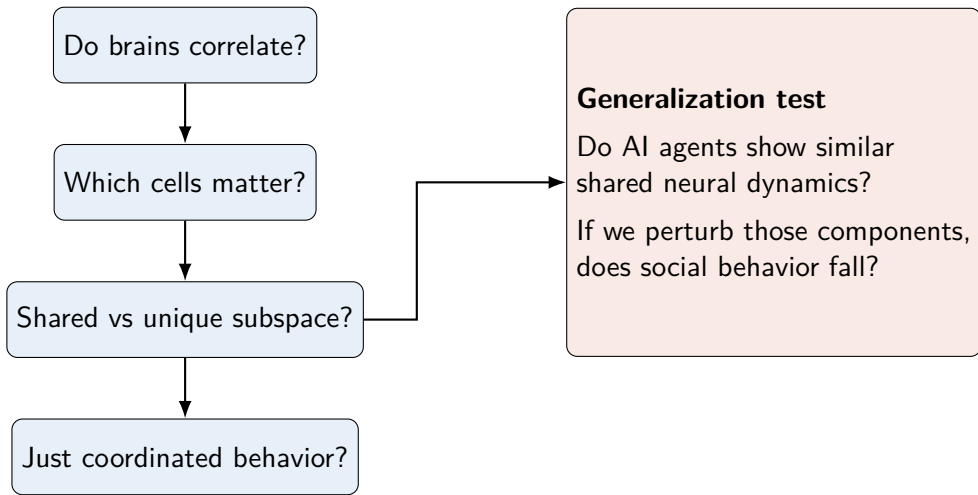


trust, competition, reciprocity, best response

Main question

When two individuals interact, do their brains develop a measurable **shared neural subspace** that cannot be reduced to the same stimulus or the same action?

- Biological side: simultaneous dmPFC calcium imaging in two mice.
- Artificial side: internal RNN activity in two multi-agent RL systems.
- Big ambition: move from **single-brain social neuroscience** to **multi-brain systems neuroscience**.



- ① Clarify the **research question** and the alternative explanations the paper tries to rule out
- ② Walk through the **experimental design** in mice and AI
- ③ Organize the **main results** into a clean story
- ④ Evaluate the authors' **interpretation**
- ⑤ Explain whether I am **convinced**
- ⑥ Propose a **better design** and a **new neuroeconomics-style extension**

Design

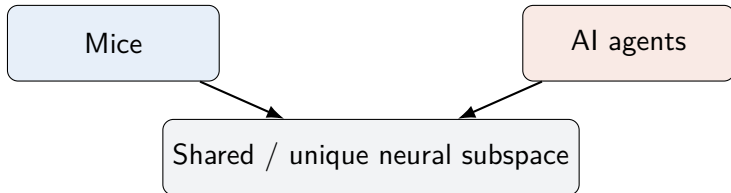
What exactly did the authors do?

Biological system

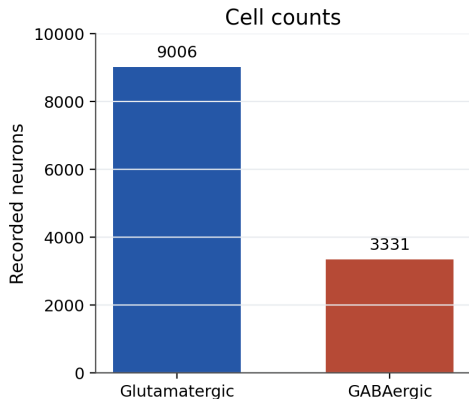
- Two freely interacting mice
- Simultaneous dmPFC calcium imaging
- Comparison of **glutamatergic** and **GABAergic** neurons
- Interaction, separation, shuffled, and cross-pair controls

Artificial system

- Two MARL agents learning to interact
- Analysis of internal RNN activations
- Comparison of **social** and **non-social** training
- Direct perturbation of shared components



- Brain region: **dmPFC**, a region linked to social information and social behavior regulation.
- Recording method: **in vivo microendoscopic calcium imaging**.
- Cell-type-specific recording:
 - CaMKII-positive glutamatergic neurons
 - mDLX-positive GABAergic neurons
- The interactions are naturalistic rather than tightly scripted trial-by-trial events.



Source: Zhang et al. (2025), *Nature*

Main condition

- **Interaction session:** two mice freely interact
- Social and non-social actions unfold naturally

Control conditions

- **Separation:** a physical divider blocks direct interaction
- **Phase-randomized:** temporal phase is scrambled
- **Cross-pair:** signals are compared across mice that never interacted

What each control tests

- Separation: not every shared environment creates coupling
- Phase-randomized: not a simple autocorrelation artifact
- Cross-pair: not any arbitrary pair of animals

Social behaviors

attack, tussle, chase, threaten, escape, defend, flinch, general sniff, face sniff, genital sniff, approach, follow

Non-social behaviors

dig, self grooming, climb, explore object, bite object, stand

Why that matters

The goal is not just to ask whether the mice are “being social.” The authors want to distinguish:

- which kinds of interaction produce stronger cross-brain coupling,
- whether non-social activity lives more in the unique subspace,
- and whether complementary actions such as *attack* versus *escape* matter more than identical actions.

Initial expectation

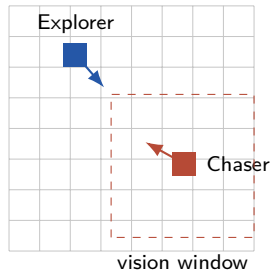
Since glutamatergic neurons dominate the cortex numerically, one could expect them to drive most population-level inter-brain correlation.

What the paper finds

The stronger social coupling appears in **GABAergic neurons**, not in the larger glutamatergic population.

This turns the paper from a descriptive synchrony result into a mechanistic question:
Which cell types carry cross-individual social information?

- Two agents: **chaser** and **explorer**
- Multi-agent reinforcement learning
- Each agent uses an **actor-critic architecture with a 256-unit RNN**
- **Social task:** partner-relevant reward and partner in vision
- **Non-social task:** social reward removed, interaction no longer behaviorally useful



Source: Zhang et al. (2025), *Nature*

Mouse side gives

- biological realism
- cell-type specificity
- natural social interaction

AI side gives

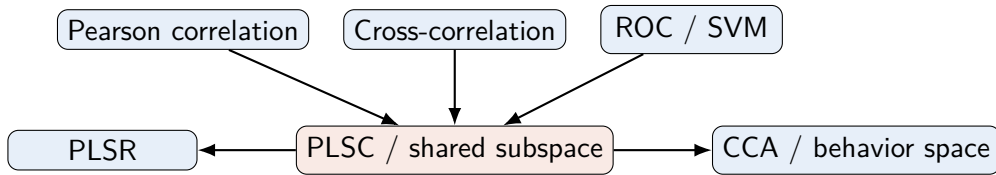
- full access to internal states
- cleaner control over rewards and inputs
- direct perturbation of the shared components

Combined logic

If shared neural dynamics appear in both systems, and perturbing them reduces social behavior in AI, the authors can argue for a more general principle of interacting intelligence.

Methods

How do they define “shared”?



- The first tools test whether cross-brain coupling exists at all.
- PLSC is the conceptual center of the paper.
- PLSR and CCA ask what kinds of behavior actually explain the shared structure.

Pearson correlation

Measure whether aggregate activity in two brains rises and falls together.

Cross-correlation

Evaluate correlation across different time lags.

Question answered

Is there inter-brain coupling above chance?

Question answered

If the peak is near 0 s, the coupling is close to synchronous rather than delayed noise.

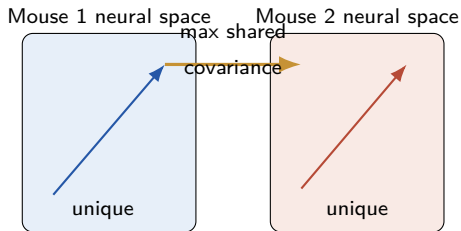
Why the controls matter

When phase randomization destroys the peak, the original cross-brain signal is unlikely to be just a by-product of one signal's internal temporal smoothness.

- Each mouse has a high-dimensional neural activity space.
- PLSC finds the dimensions that **co-vary most strongly across the two brains**.
- This is not neuron-to-neuron matching. It is pattern-to-pattern matching.

Output

Shared neural dimensions and **unique neural dimensions**



Shared neural subspace

- Activity that changes across both individuals
- More strongly linked to social interaction
- Can contain information about both self and partner

Unique neural subspace

- Activity specific to one individual
- More strongly linked to non-social or self-specific behavior
- Less tied to the partner's moment-to-moment state

Shared does **not** mean identical neurons or identical actions.
It means shared variance along specific cross-brain dimensions.

Coordinated behavioral subspace

The part of behavior that changes together across both mice.

Uncoordinated behavioral subspace

Different actions that are still linked inside the same interaction.

same action

→ approach + approach
follow + move

complementary action

→ attack + escape
chase + defend

Critical interpretation test

If the shared neural subspace can be explained entirely by coordinated behavior, then the paper's big claim about a coupled social system becomes much weaker.

- In mice, the paper mainly provides recording, decoding, and subspace analyses.
- In AI, the authors can ask a stronger causal question:

If we selectively disrupt the components that load onto the shared dimensions,
do the agents become less social?

- That is the paper's clearest evidence for **functional relevance**.

Mouse Results

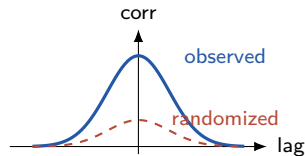
What do we learn from real brains?

- ① Inter-brain correlation appears during interaction
- ② GABAergic neurons are more socially coupled than glutamatergic neurons
- ③ Neural activity splits into shared and unique subspaces
- ④ The shared subspace is more social than the unique subspace
- ⑤ Shared dynamics are not reducible to simple coordinated movement

Main finding

During **interaction sessions**, the two mice show significant dmPFC coupling. That coupling is reduced or absent in **separation**, **cross-pair**, and **phase-randomized** controls.

- The cross-correlation peak is near 0 s.
- The signal survives pre-whitening.
- This makes a simple autocorrelation explanation less plausible.



Source: Zhang et al. (2025), *Nature*

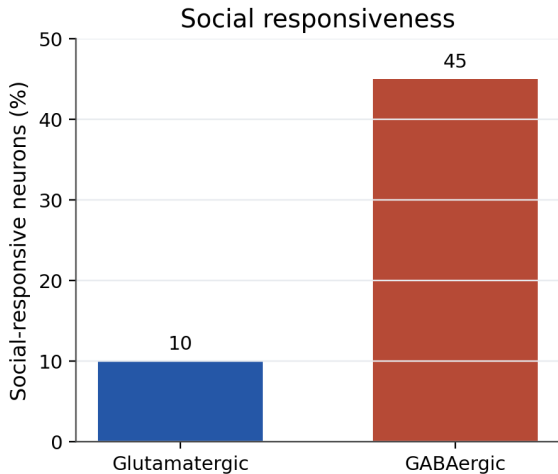
Economic intuition

In strategic interaction, choices are not independent draws.

- My best response depends on your move.
- Your move depends on what you expect from me.

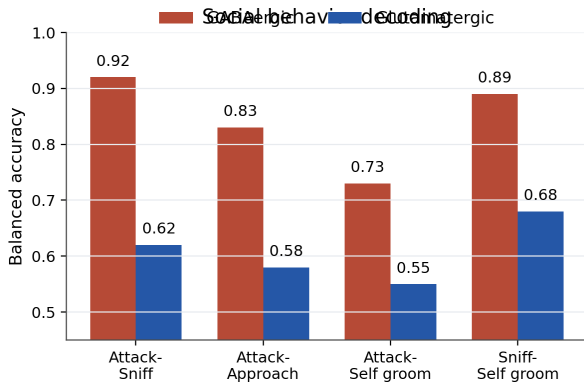
Neural translation

Inter-brain coupling suggests that neural states may also become interdependent rather than remaining isolated optimization problems.



- About **45%** of GABAergic neurons are responsive to social interaction.
- Only about **10%** of glutamatergic neurons are.
- GABAergic populations also decode social behaviors better.
- The effect is not explained by firing-rate differences.

Source: Zhang et al. (2025), *Nature*



Interpretation

GABAergic neurons are not just more correlated. They also carry richer information about social events.

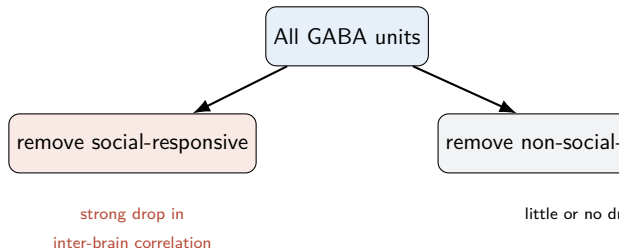
- Better social versus social classification
- Better social versus self-groom classification
- More consistent with interaction-state tracking

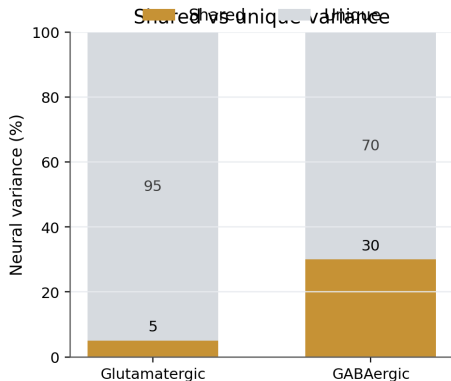
Note: bars summarize the direction and relative magnitude of the SVM decoding comparisons reported in Fig. 1q–t.

Computational removal analysis

Removing **social-responsive** GABAergic neurons causes a clear drop in inter-brain correlation.

- Removing non-social-responsive GABAergic neurons does not do the same
- Similar removals in glutamatergic populations have weaker effects



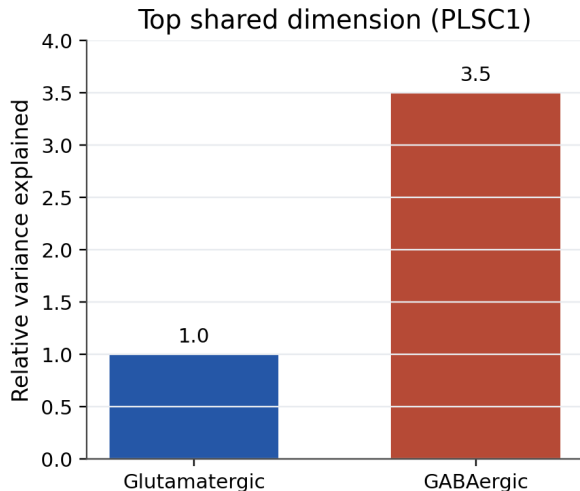


- Significant shared dimensions appear during live interaction.
- They are nearly absent in separation.
- GABAergic populations have about **30%** shared variance.
- Glutamatergic populations have only about **5%**.

Key message

The gap is not just stronger average synchrony. Much more of the GABAergic neural space changes jointly across the two animals.

Source: Zhang et al. (2025), *Nature*

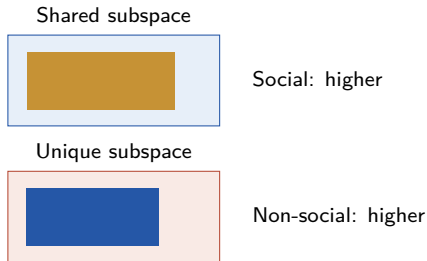


- The paper reports that GABAergic PLSC1 explains about **3.5 times** as much variance as glutamatergic PLSC1.
- So the difference is not only about having more shared dimensions.
- The leading shared dimension itself is stronger.

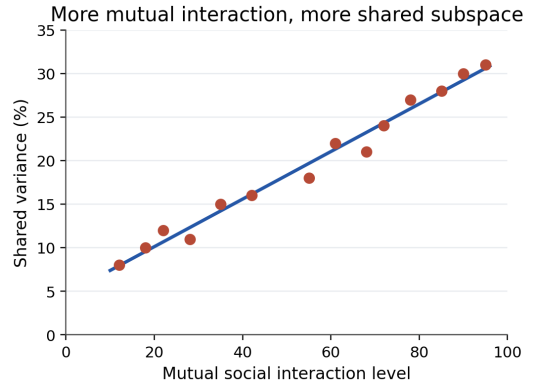
- The authors use PLSR to explain neural subspaces with behavior.
- **Social behaviors** explain more variance in the shared subspace.
- **Non-social behaviors** explain more variance in the unique subspace.

Interpretation

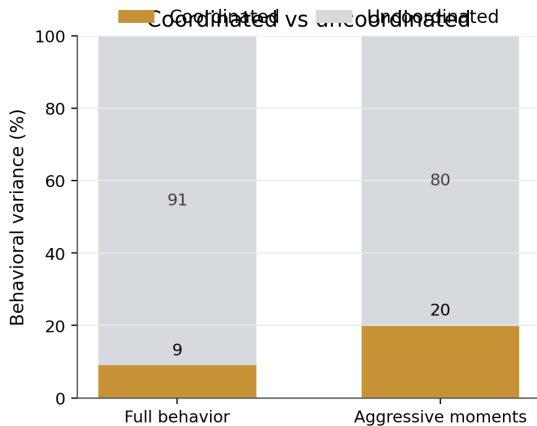
The shared subspace is not a mathematical artifact. It is behaviorally meaningful and especially tied to social interaction.



- More **mutual** social interaction is associated with more shared variance.
- Aggressive or antagonistic pairs often show especially strong coupling.
- The authors suggest that high-attention interaction may intensify cross-brain coupling.

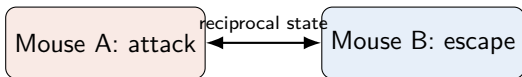


The upward trend summarizes the reported positive association between mutual social interaction and GABAergic shared variance.



- Across 27 pairs, coordinated dimensions explain only about **9%** of total behavioral variance.
- Even during aggressive interaction, coordinated dimensions explain only about **20%**.
- So the shared neural subspace also reflects **uncoordinated but socially linked** behavior.

Source: Zhang et al. (2025), *Nature*



Different actions, but one **shared interaction state**

Interpretive payoff

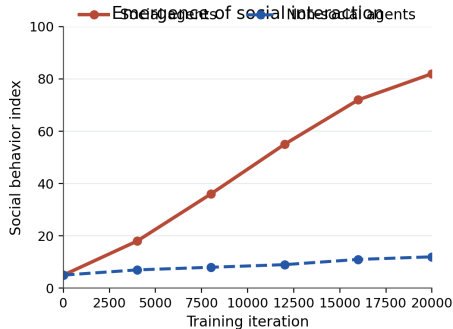
The paper is not just about behavioral synchrony. It is about a jointly represented dyadic state.

- not simple mimicry
- not same action at the same time
- more like a shared social game state

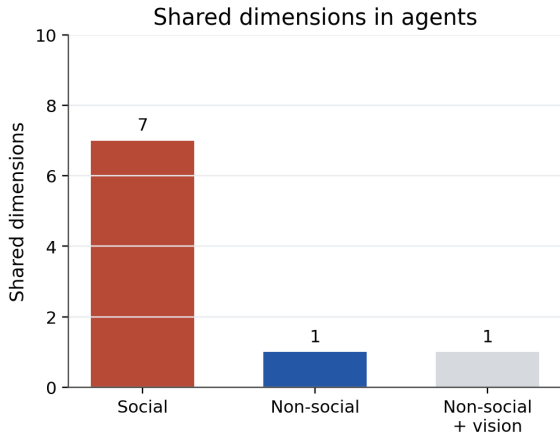
AI Results

Does the same logic appear in artificial agents?

- Socially trained chasers gain reward over training.
- Collisions increase.
- Time keeping the partner in vision also increases.
- Non-social agents learn exploration, but not the same social following strategy.



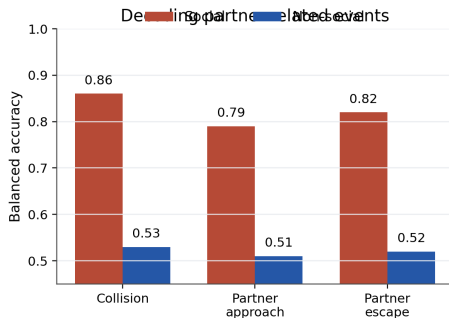
Curve shape summarizes the training pattern reported in Fig. 5c–h: social behavior rises only in the social environment.



- Trained social agents have many more shared dimensions.
- Giving non-social agents the same or even full visual input does not recreate the shared subspace.
- So shared dynamics are not just a consequence of **seeing the same world**.

Relative bar heights summarize the contrast reported in Fig. 6g and Extended Data Fig. 13a.

- Social agents' RNN activity can decode:
 - collision events,
 - partner approach,
 - partner escape.
- Non-social agents cannot do this at the same level.
- Just like in mice, **partner representation is enriched in the shared dimensions.**



What they perturb

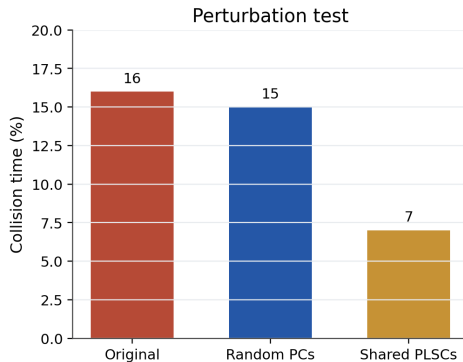
Neural components that strongly contribute to the top shared dimensions.

What happens

Social action declines, including a drop in collisions.

Why this is important

This is the paper's strongest evidence that shared components matter for behavior.



	Mice	AI agents
Shared dimensions emerge?	Yes, during live interaction	Yes, during social training
Just same input / same action?	No, behavior controls argue against that	No, shared-input and identical-action controls argue against that
Partner representation?	Yes, especially in the shared subspace	Yes, especially in social agents
Functional perturbation?	Not directly in vivo	Yes, social behavior drops after perturbation
Main limitation	Mostly correlational	Simplified task and architecture

Interpretation

Should we buy the authors' story?

Main claim

Shared neural dynamics are a **fundamental and generalizable feature** of interacting neural systems in both biological and artificial agents.

- Two interacting individuals should be treated as one integrated system.
- GABAergic neurons play a special role in the biological case.
- The shared subspace is behaviorally meaningful, not just mathematical structure.
- AI results provide functional support for the importance of shared components.

- ① **Strong controls:** separation, phase randomization, cross-pairing, pre-whitening, and behavior-space analyses
- ② **Cell-type specificity:** the story is not only “dmPFC synchronizes”
- ③ **High-dimensional framing:** the paper moves beyond average correlation
- ④ **AI perturbation:** shared components are tied to social behavior, at least in the artificial system

Causality gap in mice

The mouse evidence is still largely correlational and computational.

- We do not yet know whether shared dynamics are a cause or a consequence of social interaction.
- The paper does not directly manipulate the relevant mouse ensembles in vivo.

The AI analogy has limits

AI agents are not mouse brains.

- Different architecture
- Different learning rules
- Different motivational structure

- ① **Aggression-heavy context:** some of the strongest shared signals appear in aggressive, high-attention interaction.
- ② **One-region window:** dmPFC is only one piece of the social brain network.
- ③ **Interpretability problem:** a statistically meaningful PLSC dimension does not automatically have a clean biological meaning.
- ④ **Generalization risk:** cooperation, bonding, empathy, or teaching may show different dynamics.

I buy

- Interacting brains really do show shared structure
- GABAergic neurons seem especially important
- The effect is not reducible to simple motor coordination

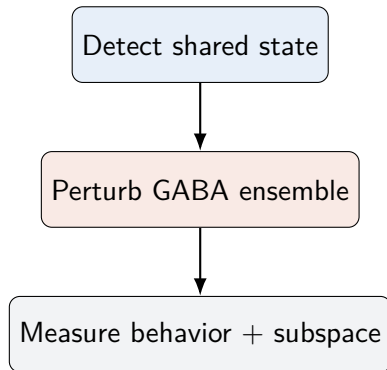
I do not fully buy

- that the paper has already proven a universal principle,
- that AI results map directly onto biological mechanism,
- or that attention, arousal, and social computation are fully separated here.

Extension

How would I redesign or extend the work?

- Target **social-responsive GABAergic ensembles** with optogenetic or chemogenetic manipulation.
- Ideally use a closed-loop design that intervenes when a high shared state is detected.
- Directly measure:
 - social interaction probability,
 - inter-brain correlation,
 - shared subspace size.



Current concern

The strongest shared signals are closely tied to aggression and close mutual attention.

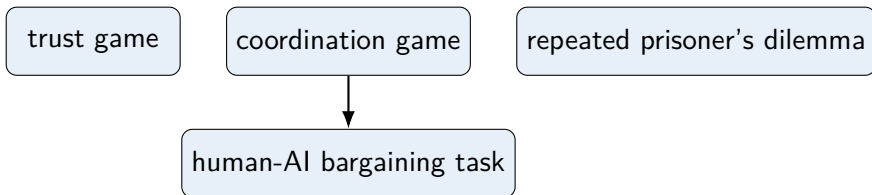
Better design

- cooperation
- mating
- parent-offspring interaction
- familiar versus unfamiliar pairs
- dominance hierarchy

Question

Is shared neural dynamics a general feature of all social interaction, or especially strong in high-arousal and high-attention interaction?

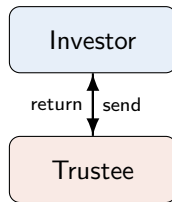
- The current mouse setting is socially rich, but not explicitly structured as a strategic game.
- To connect more directly to neuroeconomics, I would add:



Key question

Do cooperation, betrayal, trust building, and trust collapse create different shared neural states?

- Participants: human-human pairs
- Task: **repeated trust game**
- Recording: EEG hyperscanning or fNIRS hyperscanning
- Conditions:
 - human-human
 - human-AI
 - human-random-bot



hyperscanning
shared subspace
trial by trial

- ① High-trust pairs should show stronger inter-brain synchrony or shared subspace structure.
- ② As trust builds over rounds, the shared subspace should become stronger or more stable.
- ③ After betrayal, the shared subspace may weaken or reorganize.
- ④ Current shared dynamics may predict next-round cooperation.
- ⑤ Human-AI interaction may sit between human-human and random-bot conditions.

Behavioral concepts

- trust
- reciprocity
- strategic uncertainty
- belief updating
- cooperation breakdown

Neural concepts

- shared subspace
- partner representation
- coupled decision systems
- prediction of future actions

Take-home extension

Trust may not be only a preference inside one brain. It may also be a **dyadic neural process** distributed across two interacting agents.

Discussion

What should the class talk about?

- ① Are shared neural dynamics a **cause** of social interaction or a **consequence** of it?
- ② Does GABAergic dominance really reflect social coding, or could it be closer to attention or arousal gating?
- ③ Would cooperation and aggression generate different kinds of shared subspace?
- ④ How far can we compare shared dynamics in humans and AI?
- ⑤ Can the current shared state predict the next action: attack, escape, cooperate, or defect?

If you remember only one thing

The paper's biggest contribution is to treat social interaction not as one brain reacting to another individual, but as two interacting agents forming a joint neural dynamic system.

- In mice, GABAergic dmPFC neurons contain a larger shared subspace.
- Shared dynamics are not reducible to simple behavioral coordination.
- In AI, shared components emerge with interaction and matter for social action.
- For neuroeconomics, strategic interaction may be neurally **coupled**, not isolated.



Zhang, X., Phi, N., Li, Q., Gorzek, R., Zwingenberger, N., Huang, S., Zhou, J. L., Kingsbury, L., Raam, T., Wu, Y. E., Wei, D., Kao, J. C., & Hong, W. (2025).
Inter-brain neural dynamics in biological and artificial intelligence systems.
Nature, 645, 991–1001. DOI: 10.1038/s41586-025-09196-4

Appendix

Extra support slides for Q&A

- 9,006 glutamatergic neurons from 13 mouse pairs
- 3,331 GABAergic neurons from 14 mouse pairs
- Mice spent about 49% of time in active behavior
- About 48% of active behavior was social
- About 45% of GABAergic neurons were social-responsive
- About 10% of glutamatergic neurons were social-responsive
- Shared variance was roughly 30% in GABAergic populations versus 5% in glutamatergic populations

Source: Zhang et al. (2025), *Nature*

Without this control

A critic can always say:

- the animals simply move together,
- stop together,
- or enter the same high-arousal state.

With this control

The authors can argue more strongly that:

- partner-specific information enters the shared subspace,
- especially through complementary rather than identical actions.

Presentation-friendly definition

PLSC finds the cross-brain neural dimensions that **co-vary the most**, allowing the authors to separate what is **shared across two interacting agents** from what remains **unique to each individual**.

- This is a useful sentence to say if the instructor asks about the method.
- Then you can add: the shared subspace is more social, whereas the unique subspace is more self-specific.

- ① The core issue is **interactive decision systems**, not static social cue processing.
- ② Self and other are mutually endogenous, just like in strategic models.
- ③ The paper connects naturally to trust, cooperation, competition, and reciprocity.
- ④ It offers a neural version of **best-response coupling**.

- The most realistic human path is EEG or fNIRS hyperscanning, not cell-type recording.
- The main challenge is not whether synchrony exists, but whether we can build a convincing **shared versus unique subspace** analysis tied to behavior.
- Human studies are especially well suited for:
 - trust games,
 - bargaining,
 - joint problem solving,
 - human-AI collaboration.